Max Warriner

Personal Website: http://maxwarriner.github.io
LinkedIn: www.linkedin.com/in/maxwarriner
Email: mwarriner@colgate.edu
Tel: 269-769-4165

4615 Dover Hills Dr.
Apt. 205
Kalamazoo, MI 49009

Education

Colgate University, Hamilton, NY **2023-2027**
Bachelor of Arts, Applied Mathematics

- Expected Graduation Date: May 2027
- Minor: Biology
- GPA: 3.54/4.00
- Related Coursework:
  - Mathematics: Calculus I/II/III (MATH 161/162/163), Probability (MATH 316), Data Analysis I/II (MATH 354/454), Numerical Analysis (MATH 376), Real Analysis (MATH 377), Multivariate Methods for Data Analysis (MATH 4xx)
  - Biology: Bioinformatics (BIOL 316), Biostatistics (BIOL 320), Investigations in Computational Biology (MATH/BIOL 481)
- Study Abroad: University College Cork, Ireland (Spring 2026)

Honors and Awards

- Deans Award w/ Distinction: Fall 2024, Spring 2025 & Fall 2025 (Colgate University)
- Senior Mathematics Award (St. Joseph High School)

Research Experience

Computational Researcher, Dr. Bineyam Taye Lab **Feb 2025 - present**
Colgate University Department of Biology

- Researched effects of lifestyle and hygienic factors on the gut microbiome of Ethiopian school children using statistical analysis in *R*; novel discoveries in microbiome research
- Presented a poster in the spring of 2025 with other biology students; manuscript in progress

Undergraduate Thesis, Dr. Ahmet Ay Lab **Aug 2025 - Apr 2026**
Colgate University Department of Biology & Department of Mathematics

- Collaborated with biology student for undergraduate thesis
- Investigated gut microbiome & metabolome as mediator between diet and gut health in Ethiopian school children
- Presented a poster in the Fall of 2025 with other biology students; manuscript ready for submission

Research Assistant, Dr. Melissa St. Hilaire Lab **Jun - Aug 2024**
Merrimack College Department of Data Science

- Aided Dr. Melissa St. Hilaire in ongoing mathematical biology and biostatistics human research
- Optimized human circadian rhythm model in MATLAB
- Cleaned and analyze human biological data pertaining menstrual cycle, sleep, and attention using *R*

Internships

Needs Assessment Data Intern, Dr. Haleigh Leslie **May - August 2026**
Ionia County Health Department

- Assisted Dr. Haleigh Leslie (Director) with community health needs assessment by analyzing and reporting on data around Ionia County
- Generated all encompassing statistical report on community health factors

Data Science Intern, Colgate Data Science Collaboratory **Sep 2024 - Jul 2025**
Colgate University Department of Mathematics

- Programmed and developed website for the Colgate Data Science Collaboratory using *RShiny* under Dr. William Cipolli
- Performed mixed effect regression analysis for study on aging and metabolism in dogs for Dr. Ana Jimenez; presented poster for the summer

Data Science Aide, Joey Andrews **July 2024 - Aug 2025**
Michigan State House Representative - 38th District

- Compiled county and city voter data from Michigan SOS; reworked voter data project by improving efficiency and automating data mining to be completed in a fraction of time using *R*

Publications

Jiménez, Ana Gabriela, **Max Warriner**, Danny Molyneux, William Cipolli, and Alex M. Champagne. 2027. "Effects of Increases in Temperature and Genipin on Mitochondrial Function in Primary Fibroblast Cells of Dogs of Different Sizes and Ages." Biochimica Et Biophysica Acta (BBA) - Bioenergetics 1868 (1): 149605. https://doi.org/10.1016/j.bbabio.2026.149605.

Warriner, Max, Bineyam Taye. Manuscript-in-Progress. "Household and Environmental Drivers of Gut Microbiome Variation in Ethiopian School Children."

Warriner, Max*, Alta McQuillen*, Ahmet Ay, Bineyam Taye. Manuscript-to-be-Submitted. "Dietary Intake Influences GI Health Outcomes via the Gut Microbiome and Metabolome in Ethiopian Schoolchildren."

St. Hilaire, Melissa, **Max Warriner**, ... Research-in-Progress. "Optimization of parameters for the mathematical model of the effects of light on the circadian system."

St. Hilaire, Melissa, **Max Warriner**, ... Research-in-Progress. "Field-based study of menstrual cycle phase on sleep, light exposure, and performance in young women."

Programming Languages & Software

R, MATLAB, Python, LaTeX, Git